AP PRECALCULUS

Unit 1 • Worksheet 1.5B • Answer key

Worksheet 1.5B • Polynomial Functions and Complex Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which describes the graph of $p(x) = (x+2)^2(x-1)$ at $x = -2$?

(A) It has a vertical asymptote.
(B) It crosses the x-axis.
(C) It touches the x-axis without crossing it.
(D) It has a hole.

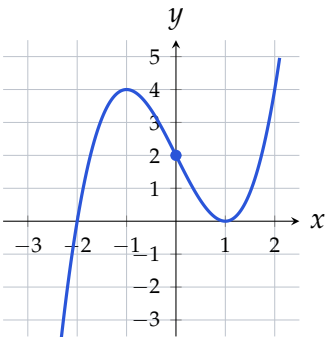
Solution

Answer: (C). Even multiplicity.

Question 2

NO CALCULATOR

The graph of a cubic q is shown; it passes through $(0, 2)$. Which could be $q(x)$?



(A) $(x+2)(x-1)^2$
(B) $(x-2)(x+1)^2$
(C) $(x+2)^2(x-1)$
(D) $-(x+2)(x-1)^2$

Solution

Answer: (A). Crosses at -2, touches at 1, rises to the right; $q(0) = 2$.

Question 3

NO CALCULATOR

At which x-intercept does the graph of $p(x) = x^3(x-2)^2(x+4)$ flatten out as it crosses the axis?

(A) $x = 0$
(B) $x = 2$
(C) $x = -4$
(D) none

Solution
Answer: (A). Multiplicity 3 (odd, at least 3).

Question 4 (Free response)

NO CALCULATOR

Let $p(x) = -(x+3)(x-1)^2(x-4)$.

(a) List the zeros with their multiplicities and describe the graph at each.
(b) Find the y-intercept and the end behavior.
(c) Sketch the graph.

Solution
(a) -3 (1, crosses), 1 (2, touches), 4 (1, crosses).
(b) $p(0) = -(3)(1)(-4) = 12$. Leading term $-x^4$: down at both ends.
(c) Down from the left, crosses at -3, up through $(0, 12)$, down to touch the axis at $x = 1$, back up, then down crossing at 4.